Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

VII SEMESTER B.TECH. (IT/CCE)

END SEM EXAMINATIONS, DECEMBER 2021

SUBJECT: PROGRAM ELECTIVE-VII INTERNET OF THINGS [ICT 4050]

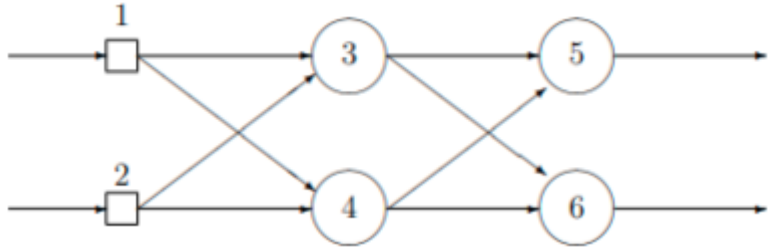
REVISED CREDIT SYSTEM
(22/12/2021)

PART B

Time: 75 Minutes MAX. MARKS: 20

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data, if any, may be suitably assumed.

1A.	With the help of a neat diagram explain IoT reference model.	**5**	**CLO3**	**AHEP:1,2,3**	**BL:2**
1B.	Write and explain a python program to gradually decrease and then increase the LED brightness every 0.1second using software PWM signal. Assume that Raspberry Pi Pin3/GPIO2 is configured as output pin and frequency of PWM is set to 500Hz.	**3**	**CLO2**	**AHEP:1,2,3**	**BL:2**
1C.	Explain any four vulnerabilities in smart grid.	**2**	**CLO5**	**AHEP:1,4**	**BL:2**
2A.	Consider the diagram Figure Q2A, represents a feed-forward neural network with one hidden layer. A weight on connection between nodes i and j is denoted by wij, such as w13 is the weight on the connection between nodes 1 and 3. The following table lists all the weights in the network.  Figure Q.2A	**5**	**CLO:4**	**AHEP: 1, 2, 3**	**BL:3**

$w_{13} = -3$ $w_{23} = 2$	$w_{35} = -1$ $w_{45} = 1$
$w_{14} = 5$ $w_{24} = -1$	$w_{36} = 1$ $w_{46} = -1$

	Each of the nodes 3, 4, 5 and 6 uses the following activation function: $\psi(v)=1$ if $v \geq 0$ otherwise 0 Here v denotes the weighted sum of a node. Each of the input nodes (1and 2) can only receive binary values (either 0 or 1). Calculate the output of the network (y5 and y6) for each of the input patterns: Pattern | P1 | P2 | P3 | P4 Node 1 | 0 | 1 | 0 | 1 Node 2 | 0 | 0 | 1 | 1				
2B.	List and explain the key identifiers of RPL network.	**3**	**CLO: 3**	**AHEP:1,2,3**	**BL:2**
2C.	Differentiate between the structured and unstructured data.	**2**	**CLO:4**	**AHEP: 1, 2, 3**	**BL:4**